Zhuoxiao Zhang

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INTERESTS

Applied machine learning and ML systems; data-driven modeling for engineering systems; scalable software for data pipelines and model deployment

EDUCATION (UM-SJTU Dual Degree Program)

University of Michigan, Ann Arbor — **Ann Arbor, MI**
B.S.E. in Computer Science | GPA: 3.94 / 4.00 — *Aug 2025 – May 2027*
Completed: EECS 370 Computer Organization; EECS 281 Data Structures & Algorithms; IOE 511 Continuous Optimization Methods (graduate); IOE 310 Introduction to Optimization Methods
Taking: EECS 445 Machine Learning; EECS 442 Computer Vision; EECS 484 Database Management System

Shanghai Jiao Tong University — **Shanghai, China**
B.S. in Electrical and Computer Engineering | GPA: 3.64 / 4.00 (Top 25%) — *Sep 2023 – Aug 2027*
Completed: ECE 3110J Electronic Circuits; ECE 2700J Logic Design; ECE 4010J Probabilistic Methods

RESEARCH EXPERIENCE

University of Michigan, Data Fusion Laboratory — **Ann Arbor, MI**
SURE Research Assistant | Advisor: Prof. Judy Jin — *Dec 2025 – Present*

- Built an end-to-end ML pipeline for lithium-ion battery health estimation from impedance (EIS) data, achieving **98.5% classification accuracy** on circuit-model structure with CNN, DNN, and MLP models in TensorFlow
- Generated simulated datasets for six candidate equivalent circuit models (ECMs) and ran self- and cross-fitting experiments to evaluate model identifiability and fitting robustness
- Developed a modular Python benchmarking suite across linear, MLP, and XGBoost model families to test whether each model's health information is consistent across model types
- Designed a weighted ranking framework with data-driven weights and robustness checks; showed that **the best-fitting model is not the best predictor**, motivating task-aware model selection

PROJECT EXPERIENCE

Isuzu Motors × University of Michigan MDP — Battery-Electric Fleet Routing — *Jan 2026 – Present*

- Developed routing optimization algorithms for an industry-sponsored battery-electric fleet planning system, implementing a **TSP baseline** and integrating **Google OR-Tools** for constrained vehicle routing with a physics-informed energy model
- Designed a **dynamic replanning** approach incorporating real-time traffic updates, route stability, and operational constraints to adapt existing routes to changing conditions
- Integrated optimization algorithms into a **Flask**-based backend through **REST APIs**, collaborating with frontend teammates to deliver an interactive route planning application

Battery SOH Prediction & Explainability Dashboard | Live Demo — *Personal Project*

- Trained XGBoost and LSTM SOH regressors on the HUST dataset (**77 cells, 146k cycles**) with a cell-level train/test split; LSTM reached **MAE 0.0070 / RMSE 0.0094** (XGBoost MAE 0.0174)
- Deployed a **Dockerized Streamlit app on Hugging Face Spaces** with degradation visualization, end-of-life estimation, and SHAP explainability

ColorVision++ | GitHub — *Personal Project*

- Built a real-time color analysis app (**C++17, OpenCV**) featuring perceptual color recognition, K-Means++ dominant color extraction, and colorblindness simulation
- Implemented and benchmarked a **CIEDE2000-based color classifier** against an HSV rule-based baseline, improving broad-category accuracy from **64% to 84%** on a small color dataset

Spotify Lyrics Viewer | GitHub — *Personal Project*

- Built a desktop app (**Electron, JavaScript**) that shows lyrics for the currently playing Spotify track, handling Spotify OAuth, lyrics sourcing via the NetEase API, and an always-on-top overlay window
- Extending with Whisper-based forced alignment, LLM-based lyrics disambiguation, and streaming translation; shipping auto-update releases for daily use

TEACHING & OUTREACH

Teaching Assistant, Ethics of AI, Shanghai Jiao Tong University Feb 2025 – Apr 2025

- Supported course operations by managing attendance records, organizing learning materials, and communicating course announcements to students.

Volunteer Teacher, Yangjia Middle School, Hunan Dec 2023

- Volunteered at a rural middle school in Hunan, introducing students to mathematical and computing concepts through engaging, accessible lessons.
- Helped broaden students' exposure to STEM subjects and encouraged curiosity and confidence in learning.

HONORS & CREDENTIALS

First Prize, National Olympiad in Informatics in Provinces (NOIP), Shanghai (2021)
Accenture IOE Scholarship (2026, selected recipient); Michigan Engineering Continuing Student Scholarship (2026)
SJTU Undergraduate Excellence Scholarship (2024, 2025)
Tests: GRE 328 (Q170 / V158 / AWA 4.5); TOEFL iBT 6.0 (R5.5 / L6.0 / S5.5 / W6.0)

SKILLS

Languages: C++, Python, Java, C, JavaScript, SQL, MATLAB, HTML/CSS
Frameworks: TensorFlow, scikit-learn, XGBoost, LightGBM, React, Electron, OpenCV, Streamlit
Tools: Git, Docker, Snowflake, Google OR-Tools, Hugging Face Spaces
ML: CNN, LSTM, gradient boosting, SHAP, feature engineering, time-series analysis